

ORBIT PULSE — CREW HEALTH CONSOLE

Complete Screen Recording Actions, On-Screen Captions & Narration Script

PROJECT TYPE

Space Health Web App

TARGET DURATION

120 Seconds (2:00 Min)

RESEARCH DATASET

NASA OSDR Cache v1.0.0

CORE CONCEPT

Non-Diagnostic Monitoring

0:00 - 0:15

Problem Hook

0:15 - 0:40

Vitals & Replay

0:40 - 1:05

Multi-Signal Alert

1:05 - 1:25

OSDR & Safety Guard

1:25 - 1:45

Timeline & Baselines

1:45 - 2:00

Mission Vision

Scene 1: The Problem & Solution Hook

0:00 - 0:15 (15s)

SCREEN RECORDING ACTIONS

- Open Dashboard:
`http://localhost:3000/dashboard`
- Press **F11** for fullscreen browser view.
- Highlight the animated deep-space canvas, glowing title, and prominent **SIMULATED DATA** badge.

ON-SCREEN CAPTIONS

ORBIT PULSE — CREW HEALTH CONSOLE

NASA Space Apps Challenge

ENGLISH VOICEOVER SCRIPT

"During long-duration space missions, microgravity, radiation, and confinement alter human physiology in subtle, interconnected ways. Traditional systems either apply rigid Earth-bound ranges or attempt dangerous medical diagnoses. Introducing Orbit Pulse — a crew health monitoring console designed to track personal biometric deviations without claiming medical diagnosis."

বাংলা গাইড: শান্ত কিন্তু গভীর ও আত্মবিশ্বাসী গলায় শুরু করুন। মহাকাশের কঠিন বাস্তবতা এবং অ্যাপটির লক্ষ্য স্পষ্ট করুন।

Scene 2: Live Telemetry & Deterministic Replay Engine

0:15 - 0:40 (25s)

SCREEN RECORDING ACTIONS

- Hover over the **6 Vital Cards** (Heart Rate, HRV, SpO2, Sleep, Core Temp, Exercise).
- In **Replay Controls**, click **Play**, switch speed to **2x / 5x**, and gently scrub through the mission time slider.
- Click signal tabs (Heart Rate, HRV) to showcase the dynamic shaded baseline band (Mean $\pm$ 2 σ).

ON-SCREEN CAPTIONS

6 Monitored Biometric Signals

Deterministic Replay Engine (1Hz Telemetry)

Rolling Baseline Bands

ENGLISH VOICEOVER SCRIPT

"At its heart, Orbit Pulse continuously processes six vital signals. Instead of arbitrary thresholds, the system computes personal rolling baselines with z-score anomaly detection. With our deterministic replay engine, mission flight surgeons and astronauts can scrub backward or fast-forward through time, observing deviations against historical baseline bands in real time."

বাংলা গাইড: মাউস কন্ট্রোল খুব স্মুথ রাখুন। স্ক্রাবার নাড়ানোর সাথে সাথে চার্টের আপডেটের ছন্দ ভয়েসের সাথে ম্যাচ করুন।

Scene 3: Anomaly Event & Automatic Fullscreen Alert

0:40 - 1:05 (25s)

SCREEN RECORDING ACTIONS

- Scrub replay to **07:30 (Index 450)** — the predefined anomaly trigger point.
- The app automatically pops into Fullscreen Multi-Signal Alert modal!**
- Point cursor to the deviating signals (e.g. Heart Rate spike + HRV drop), baseline values, and z-score breakdown.
- Press **Esc** to return to dashboard; observe the glowing red **MULTI-SIGNAL** cards.

ON-SCREEN CAPTIONS

CRITICAL: Multi-Signal Deviation Detected

Automated Emergency Fullscreen Focus

ENGLISH VOICEOVER SCRIPT

"When correlated biometric shifts happen simultaneously — such as an elevated heart rate combined with an HRV drop within 48 hours — Orbit Pulse immediately launches an automated fullscreen alert. It cuts through cockpit cognitive overload, spotlighting exactly what shifted, the statistical magnitude, and personal baseline differences."

Judge Impact Moment: স্বয়ংক্রিয় ফুলস্ক্রিন অ্যালার্ট জাজদের সবচেয়ে বেশি দৃষ্টি আকর্ষণ করবে। এটি মিস করবেন না!

Scene 4: Transparent Explanations & NASA OSDR Research

1:05 - 1:25 (20s)

SCREEN RECORDING ACTIONS

- Navigate to /alerts and click on the recent alert.
- Click **"Explain this alert"** button. Show the plain-language summary and the **"Safety check: passed"** badge.
- Scroll down to **"Related NASA research"** showing real cached OSDR space studies (e.g. OSD-120).
- Open /research page and type "microgravity" in search bar.

ON-SCREEN CAPTIONS

Plain-Language Explanation Engine

Medical Safety Guardrails (Zero Unverified Claims)

NASA OSDR Study Integration

ENGLISH VOICEOVER SCRIPT

"Orbit Pulse strictly enforces medical safety principles. Rather than diagnosing illnesses, our explanation layer translates mathematical anomalies into plain, transparent language. Crucially, each alert connects directly to NASA Open Science Data Repository studies, granting crew and medical support immediate access to peer-reviewed spaceflight biological data."

বাংলা গাইড: NASA Open Science Data Repository (OSDR) নামটি পরিষ্কার উচ্চারণে বলুন, এটি নাসার মূল জাজিং ক্রাইটেরিয়ার সাথে জড়িত।

Scene 5: Mission Timeline, Crew Profile & Settings

1:25 - 1:45 (20s)

SCREEN RECORDING ACTIONS

- Go to /timeline: Showcase the mission-day dot progression (Normal, Warning, Multi-Signal days).
- Go to /crew: Display Astronaut CREW-01 overview and personalized baseline statistics.
- Go to /settings: Drag the Z-Score threshold slider (2.0 to 5.0) to demonstrate dynamic operator control.

ON-SCREEN CAPTIONS

Mission Timeline Progression

Personalized Baseline Matrix

Dynamic Z-Score Threshold Slider

ENGLISH VOICEOVER SCRIPT

"The platform provides complete longitudinal awareness with its Mission Timeline and comprehensive Crew Profile. Mission control operators can adjust z-score thresholds and temporal windows in settings, calibrating sensitivity whether the crew is sleeping, conducting an EVA, or transitioning orbits."

বাংলা গাইড: প্রতিটি পেজে ১-২ সেকেন্ড মাউস থামিয়ে দ্রুত ব্রাউজ করুন যাতে অ্যাপের গভীরতা বোঝা যায়।

SCREEN RECORDING ACTIONS

- Return to main /dashboard with the pulsing telemetry and space backdrop.
- End with a closing title card displaying: Team Name, Challenge Name, and GitHub link.

ON-SCREEN CAPTIONS

Orbit Pulse — Protecting Humanity Beyond Earth

NASA Space Apps Challenge 2026

ENGLISH VOICEOVER SCRIPT

"By harmonizing personal baseline telemetry with NASA open science research, Orbit Pulse ensures that wherever humanity ventures into the cosmos, astronaut health remains transparent, protected, and empowered. Thank you."

⚡ Quick Alternative: 30-Second Elevator Pitch Script

(যদি গাইডলাইনে ৩০ সেকেন্ডের টিজার বা শর্ট ভিডিও চাওয়া হয়)

0:00 - 0:08 (Dashboard): "In deep space, subtle biometric shifts can signal mission-critical physiological changes. Meet Orbit Pulse."

0:08 - 0:16 (Auto Fullscreen Alert at 07:30): "Using personalized baseline anomaly detection, our console spots multi-signal deviations and automatically triggers an emergency focus HUD."

0:16 - 0:24 (OSDR & Explain): "It delivers safe, non-diagnostic explanations backed by NASA Open Science Data Repository research."

0:24 - 0:30 (Outro): "Orbit Pulse: Safeguarding astronaut wellness for humanity's next giant leap."

📹 Video Recording Quality

Record in 1080p (1920×1080) at 60 FPS using OBS Studio or Windows Game Bar (Win + G).

🖥️ Browser Setup

Press F11 to hide tabs/bookmarks. Use dark browser theme and ensure 100% zoom.

🎙️ Audio & Voiceover

Use a noise-canceling mic or AI voice (ElevenLabs / CapCut Text-to-Speech) for clear English diction.

🚀 Submission Link

Upload to YouTube as *Unlisted* or *Public* with 1080p resolution and add GitHub repo link.